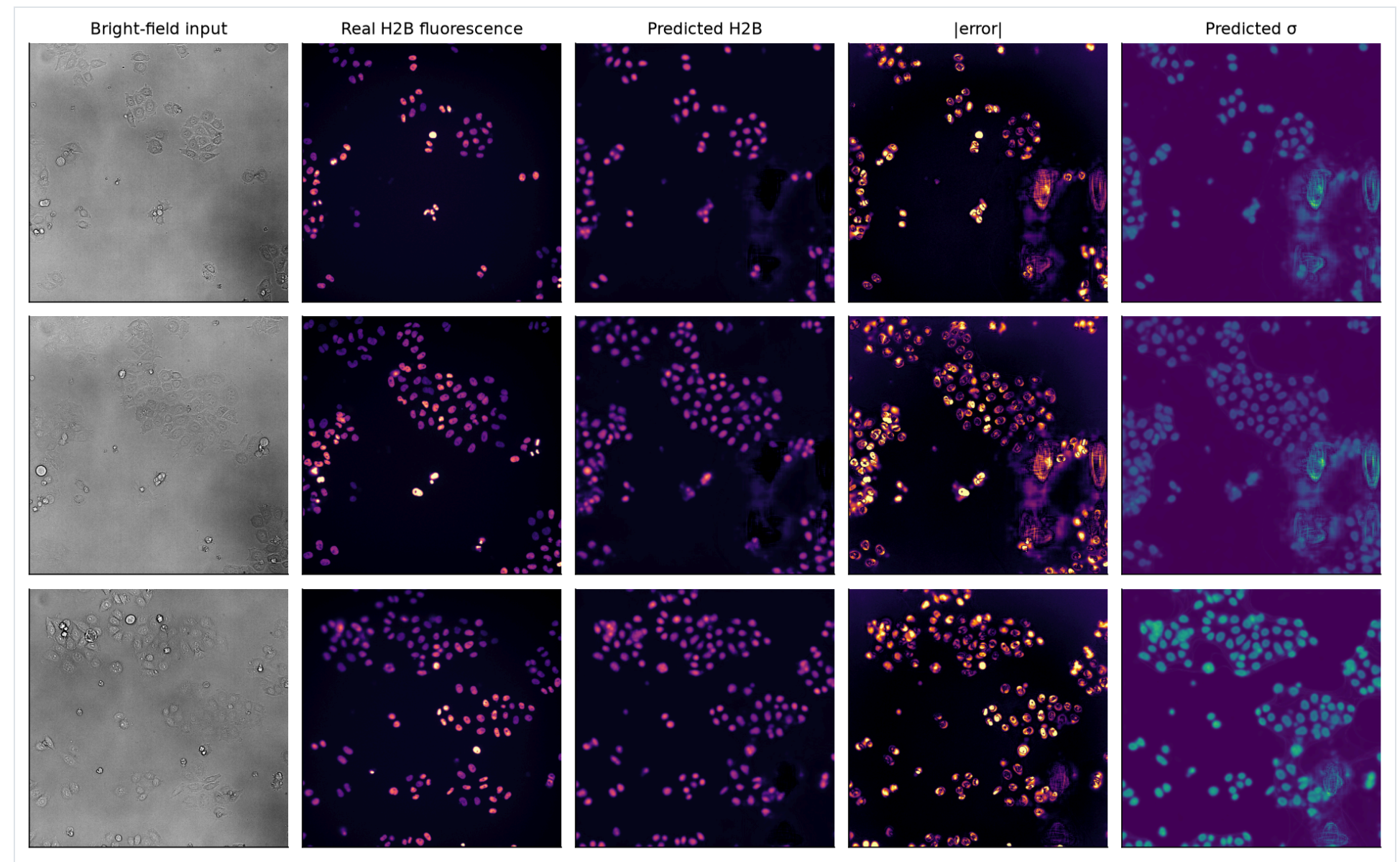


ChipStain

Label-free nuclear staining that tells you where not to trust it

- Team Hack2Publish — Md. Hamid Hosen, Esfer Sami, Foysal (Computer Science and Engineering)



Nuclear read-outs need a stain — and a stain has a cost

- Cell number, growth and viability on an organ-on-a-chip all start from a nuclear stain
- Fixation ends the experiment; live DNA dyes are phototoxic
- Reporter lines are rarely available for iPSC-derived tissue such as neurons
- In-silico labeling predicts the stain — but a point prediction gives no warning when it is wrong

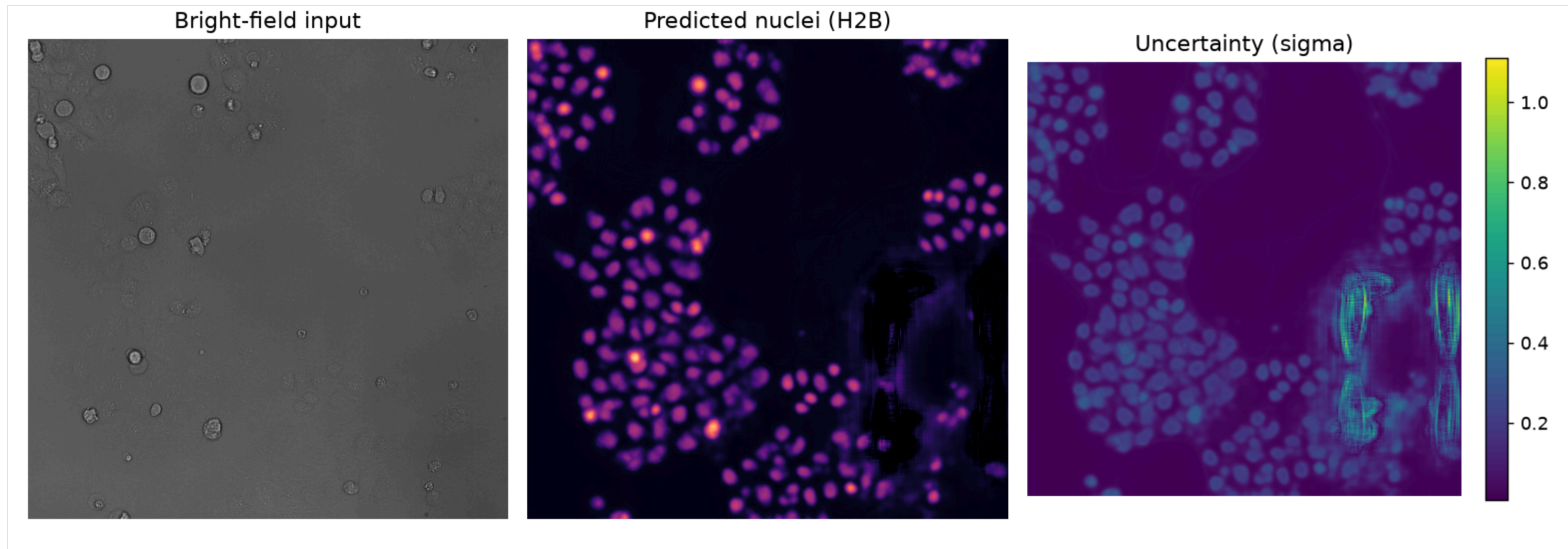
Predict the stain and its uncertainty



- U-Net, ImageNet ResNet-34 encoder; head predicts mean μ and variance σ^2 per pixel
- β -NLL loss keeps the variance from swallowing hard pixels
- 8× test-time augmentation: $\sigma^2 = \text{learned variance} + \text{TTA disagreement}$
- No new mechanism — the contribution is testing whether σ actually helps

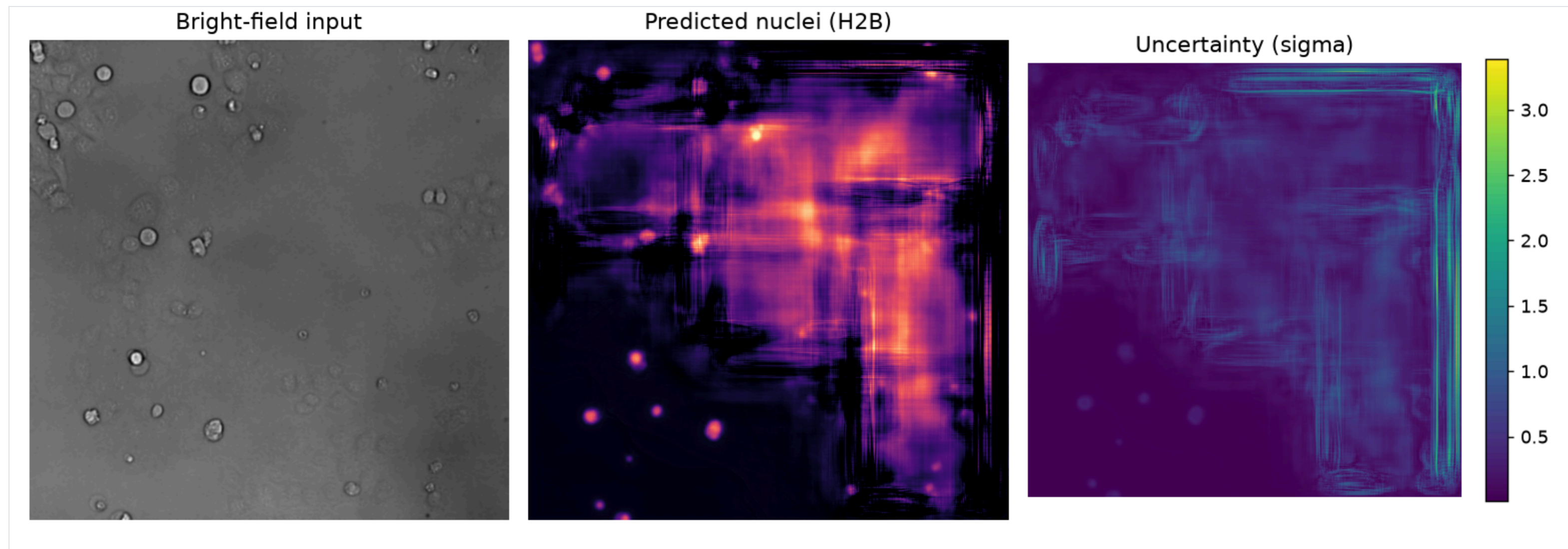
183 nuclei found — the reference has 180

- Bright-field in → predicted H2B and σ out, on a laptop CPU
- σ is higher on nuclei and near zero on background
- Mean $\sigma = 0.08$



The prediction fails — this model's σ says so

- Mean σ jumps from 0.08 to 0.33 and the app warns
- Honest caveat: across 3 training runs, σ rises under blur in only 1 of 3 — the same is true of the baselines
- So σ is not a reliable drift alarm; check new imaging with paired images



Same fidelity, much better uncertainty

Pearson r

0.768 vs 0.764

$\rho(\sigma, \text{error})$

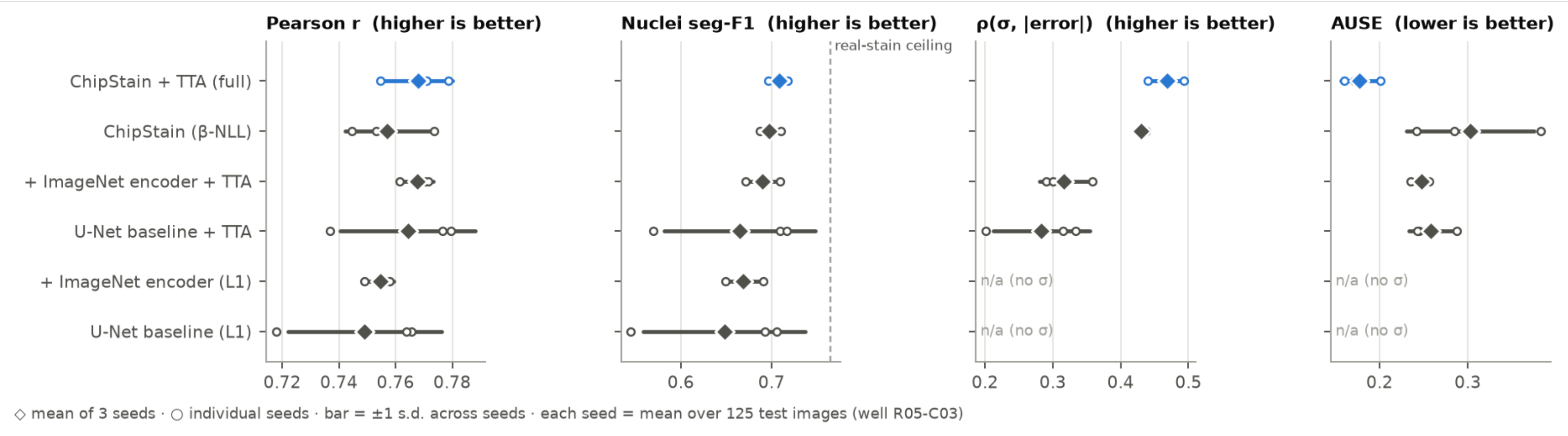
0.47 vs 0.28

95 % interval coverage

92 %

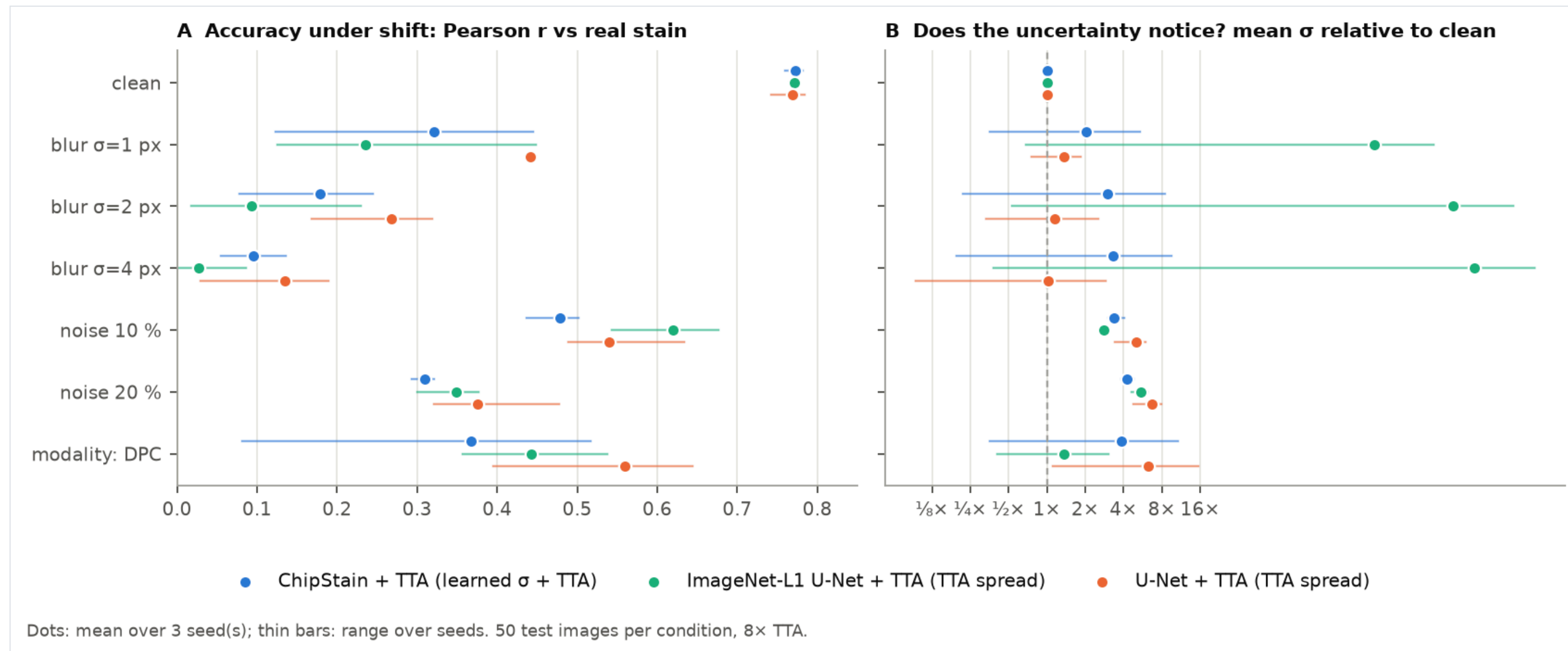
MAE cost

+10 %



Under imaging shift, σ is not a reliable alarm

- Blur and phase-contrast input break every model's prediction
- Whether σ rises depends on the training run, for every model (ChipStain: 1 of 3 seeds)
- Added noise is flagged by every model in every seed



60 hours of growth from bright-field alone

doubling time, label-free

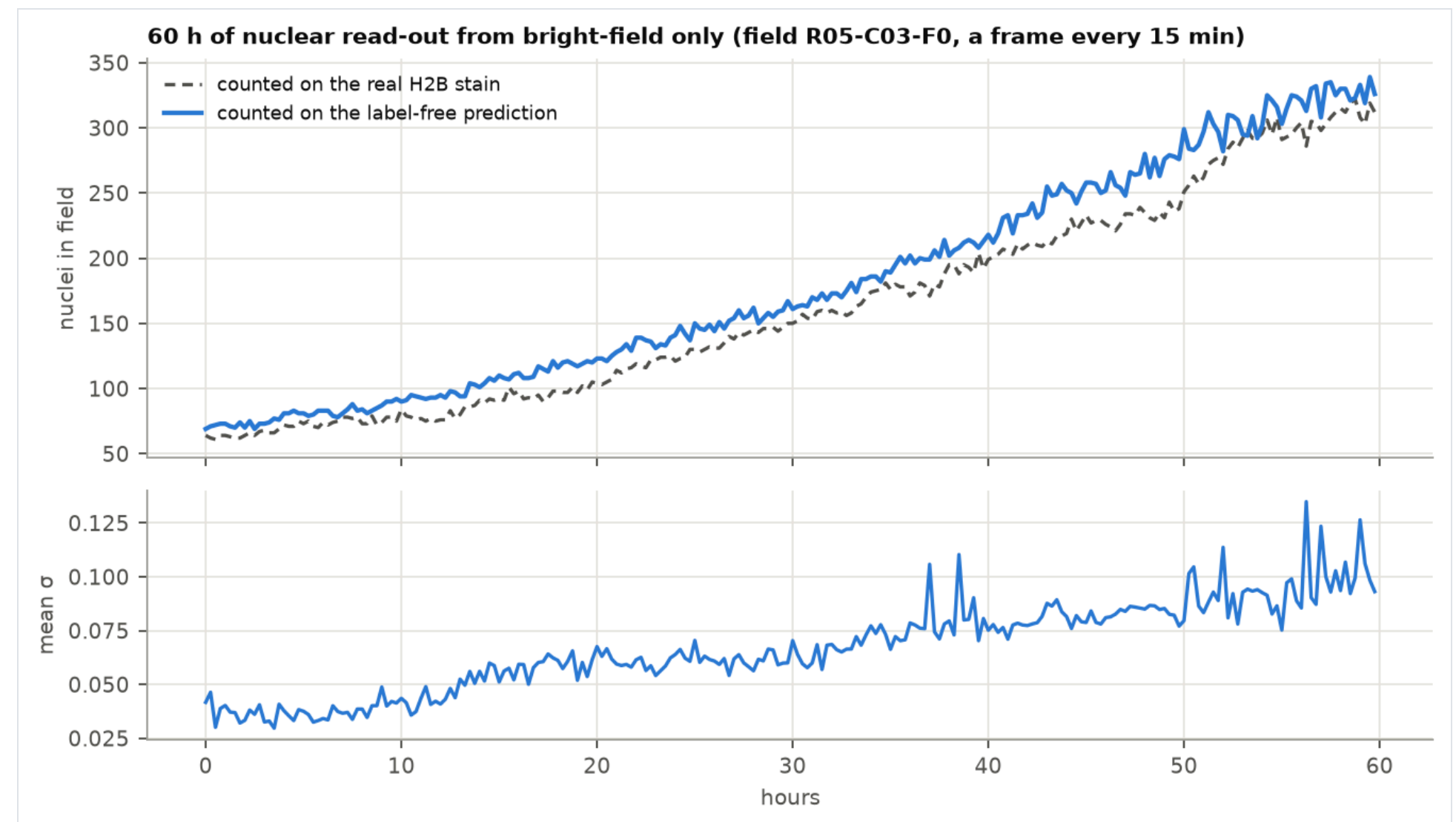
24.9 h

from the real stain

23.9 h

error without $\rightarrow$ with σ gate

9.1 $\rightarrow$ 4.1 %



What σ does not do

- For counting alone, direct segmentation wins: Cellpose F1 0.831 vs ours 0.708
- Per-nucleus σ is no better than a simple brightness rule
- No organ-on-chip images were available — validation is in a plate and a dish
- Neurons (iPSC-derived, in a dish): zero-shot r 0.59 $\rightarrow$ 0.76 after fine-tuning on one well

github.com/hamidhosen42/AI4S-Open-Innovation-AI-for-Life-Science

- Code, seed-0 weights, data links, CPU inference script
- One script regenerates every number, figure, this deck and the report
- Team Hack2Publish — Md. Hamid Hosen · Esfer Sami · Foysal (Computer Science and Engineering)